Experimental investigation into the thermomechanical vibratory response of uncooled rotating detonation engine combustor liners

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I. Abstract

Rotating Detonation Engines (RDEs) have the ability to revolutionize the propulsion and energy industries with their theoretical high efficiencies compared to traditional deflagration-based systems and reduced mechanical complexity. The transition from theory to operational hardware has proved to be challenging. The primary road blocks are the incomplete understanding of structural failure mechanisms and optimal operating conditions.

Of particular concern is the potential for high-cycle fatigue (HCF), a damage mechanism that has historically been responsible for numerous failures in aerospace systems. Each detonation wave passage constitutes a loading cycle, and with operating frequencies on the order of one to tens of kHz, an RDE can accumulate millions of cycles within tens of seconds. The severity and importance of HCF in RDEs, however, remains poorly characterized. While some analytical work has been conducted investigating HCF and dynamic amplification in cooled RDEs [1, 2], experimental validation is lacking. Teasly has reported the destruction of interfaces, bolts backing out, and the shattering of metal seals – all potentially indicating unbalanced detonation wave loading and vibrational phenomena [3].

Initial Finite Element Analysis (FEA) completed by the authors for uncooled RDEs, similar to the 3-inch MVP [4] (Figure 1), suggests that natural frequencies of an RDE (inner or outer) body reside within and dangerously close to typical RDE operating frequency ranges. Figure 1 (left) shows the particular mini-RDE of interest in operation from the University of California, Irvine [5]. Figure 1 (right) illustrates some potential mode shapes the RDE experiences in its operating frequency range. This proximity suggests that dynamic amplification and resonance conditions cannot be dismissed and need further investigation. When near a resonance condition, the dynamic amplification can dramatically increase stress levels, accelerating fatigue damage accumulation and therefore reducing the RDE's life.

The goal of this manuscript is to further experimentally investigate vibrational phenomena in RDE combustor liners, with emphasis on identifying resonance conditions and characterizing their impact on materials degradation and structural integrity. This will be done through a combination of FEA, experiments with high-fidelity diagnostic instrumentation (e.g., Laser Doppler Vibrometry), and pre/post test microscopy studies. This work aims to:

- Identify the first material microstructural evidence of resonance amplification.
- Validate FEA-based predictions of natural frequencies and mode shapes in RDE liners.
- Experimentally demonstrate the occurrence of resonance conditions at the structural scale.
- Quantify the magnitude of dynamic amplification under different operating frequencies.
- Provide design guidance for avoiding and mitigating resonance in future RDE development.

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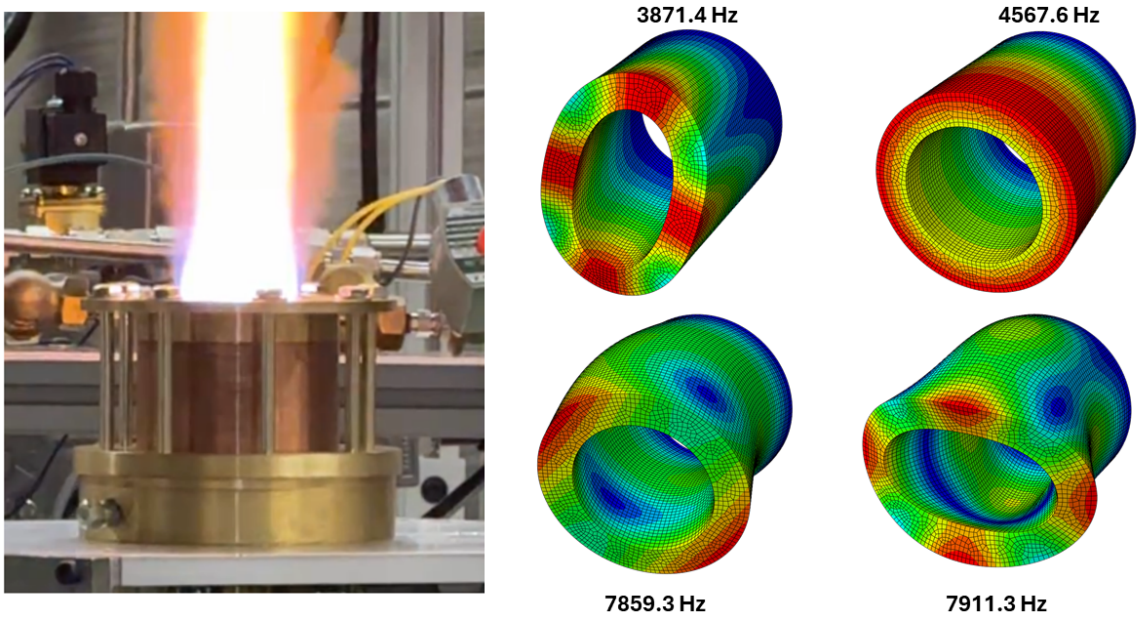


Fig. 1 (Left) University of California, Irvine 2-inch diameter mini-RDE mid burn. (Right) Mode shapes and the associated natural frequencies of the RDE outer body. The color gradient is for illustrative purposes; red corresponds to high displacement and blue to minimal displacement.

These goals collectively represent a systematic experimental framework for characterizing resonance response in RDEs. The results are intended to inform the wider RDE development community on the importance of resonance.

A 2-inch diameter mini-RDE [5] has be constructed and tested at the University of California, Irvine. The mini-RDE was designed for high throughput testing and multi-second burns. Despite the small size, the mini-RDE exhibits the same fundamental physics and loading characteristics as larger RDEs [6]. Copper will be used as the outer body and will be where analysis is focused. The copper will be manufactured via Laser Powder Bed Fusion and Cold Spray in order to investigate initial material microstructure effects. Copper has been used previously in RDEs and is a good candidate for fundamental materials studies [7, 8].

The detonation wave frequency (loading frequency) is primarily controlled by the mass flow rate of propellants and the mixture ratios between fuel and oxidizer, which are in this case methane and oxygen. Changing the flow rate and equicalence ratio will be the main methods for adjusting the loading frequency and targeting resonance conditions. These conditions will be identified using the preliminary FEA-based results. The primary diagnostic tool for this investigation is a Laser Doppler Vibrometer (LDV), a non-contact optical instrument capable of measuring the surface velocity and frequency response of an RDE. The LDV directs a laser beam (which can be modified using a beam splitter) at the target surface, the outer body of the RDE, and analyzes the Doppler shift of the reflected light. The outer body is chosen at first for this study because of the harsh optical environment RDEs present, in particular (i) flame luminosity (ii) gas density gradients and (iii) particulate matter.

To address any additional optical challenges inherent to the RDE environment, the LDV system will be configured with appropriate bandpass filtering and signal conditioning to suppress contamination. A beam splitter configuration will enable quasi-simultaneous measurements at multiple axial and circumferential locations on the outer body surface, allowing the spatial character of vibrational mode shapes to be reconstructed experimentally and compared directly against FEA-based predictions. Simultaneous high-speed pressure measurements will allow for direct correlation between detonation wave passage and dynamic structural response. The target frequency resolution of the LDV acquisition system will be sufficient to resolve individual spectral peaks within the 10–40 kHz operating range, ensuring that resonance conditions can be identified and distinguished from adjacent non-resonant operating points. As an additional method for verifying detonation and measuring the velocity of the detonation wave, a high speed camera is

set up to view down the combustion annulus. A mirror is placed down stream of the RDE and angled at the high speed camera. This set up allows for viewing of the detonation environment while keeping the high speed camera safe from the harsh RDE thrust jet.

During a RDE burn the temperature of the copper specimen (outer body) will dramatically increase. The effects of temperature on the material properties of copper cannot be ignored. Thermocouples will be installed at strategic locations to monitor the material temperature and thermal gradient information. Post-burn correlation studies between FEA and experimental measurements will be used to adjust the predicted natural frequencies.

Throughout a test series (multiple burns) the outer body will be periodically, non-destructively imaged in order to track the evolution of apparent surface damage. Additionally, measurements with a digital micrometer and a laser confocal microscope will also be made to to learn about potential shape changes due to erosion, oxidation, and/or melting and the evolution of surface roughness. This imaging history will inform the removal of a variety of cross-sectional and longitudinal cut specimens. These specimens, corresponding with optically damaged and undamaged regions will be examined via Scanning Electron Microscope (SEM) and Energy Dispersive Spectroscopy (EDS). The surface morphology will be analyzed at high magnification and elemental species identification will be made. Some of the indicators that would suggest fatigue damage are (i) fatigue crack initiation sites at the surface (ii) crack propagation features (striation patterns) (iii) grain boundary damage and (iv) slip band formation indicating plastic deformation. Particular attention will be paid to regions predicted by FEA that experience maximum stress during vibrational modes. The detected microstructural features will be compared with those predicted by calculation of phase diagrams (CALPHAD)-based simulations. Modified processing pathways will be suggested to control both detrimental and advantagous features.

In summary this paper will provide the first experimental validation of FEA-based dynamic amplification identification. It will also give insight into the importance and severity of resonance in uncooled RDEs.

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